

Probability Theory & Applications



Prithwijit Guha
Dept. of EEE, IIT Guwahati

Overview

- Introducing Probability: Basic Definitions
- Concept of Random Variables
- Probability Distributions
- Conditional Probability, Independence and Bayes Theorem
- Mathematical Expectation
- Chebyshev Inequality

Probability: Basic Definitions

An experiment is performed n times where a particular outcome **A** is observed to appear n_A times. Then, the relative frequency $\frac{n_A}{n}$ approaches the probability $P(A)$ of outcome **A** as the experiment is performed a large number of times.

$$\lim_{n \rightarrow \infty} \frac{n_A}{n} = P(A)$$



T H T T T H H H T T H T H T T H T T T H H T T T H T H T T T H T H

Probability: Basic Postulates

Probability of
Impossible Event A_I

$$P(A_I) = 0$$

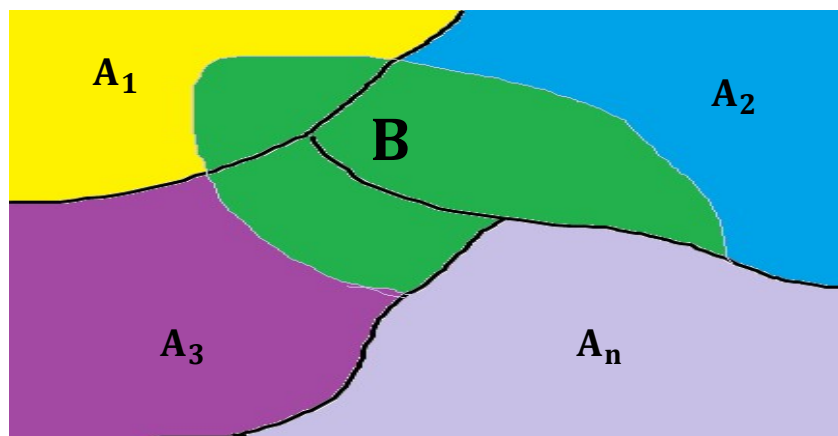
Probability of
Certain Event A_C

$$P(A_C) = 1$$

$$0 \leq P(A) \leq 1$$

Probability of any Event A

Universal Set of Mutually Exclusive Events

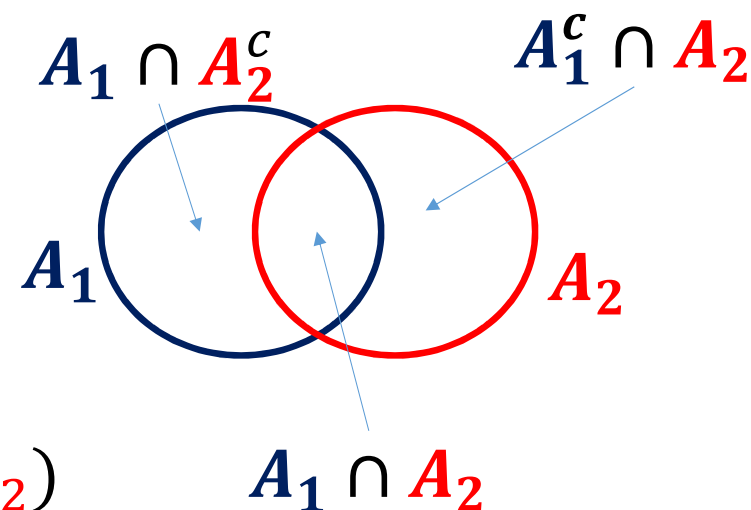


$$U = \bigcup_{i=1}^n A_i \quad \longrightarrow \quad \sum_{i=1}^n P(A_i) = 1$$

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$



$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$



Concept of Random Variables

Consider an experiment having n possible outcomes $A_1, \dots, A_n$ with respective probabilities $P(A_1), \dots, P(A_n)$. A Random Variable X maps each outcome A_i to a number $X_i = X(A_i)$ which in turn is associated with the probability $P(A_i)$.



T H T T T H H H T T H T H T T T H H T T T H T H T T T H T H

$$X_1 = X(T) = 0; P(T) = 0.61$$

$$X_2 = X(H) = 1; P(H) = 0.39$$

Distribution and Density Functions

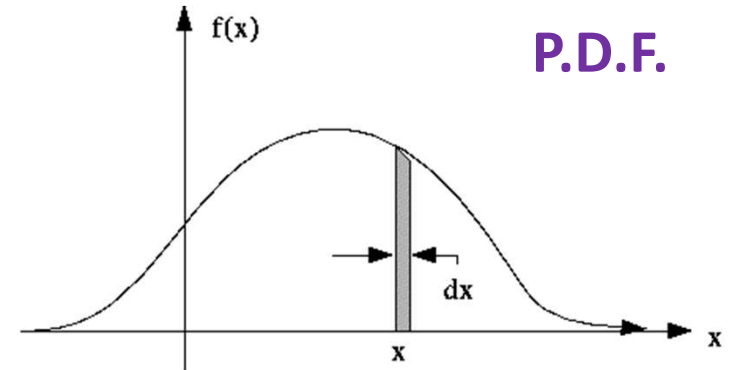
Probability Distribution Function

$$F_X(x) = P\{X \leq x\}$$

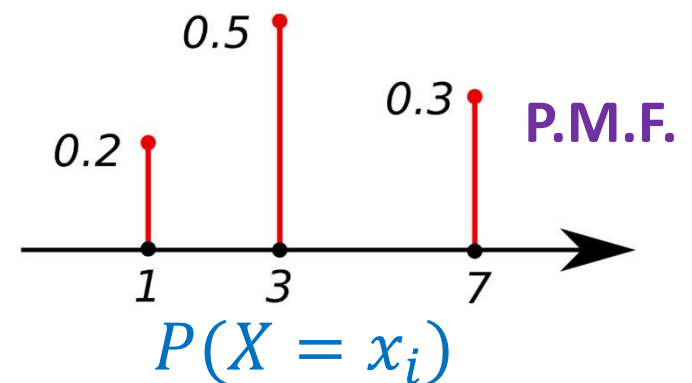
$$x_2 > x_1 \rightarrow F_X(x_2) \geq F_X(x_1)$$

Probability Density Function

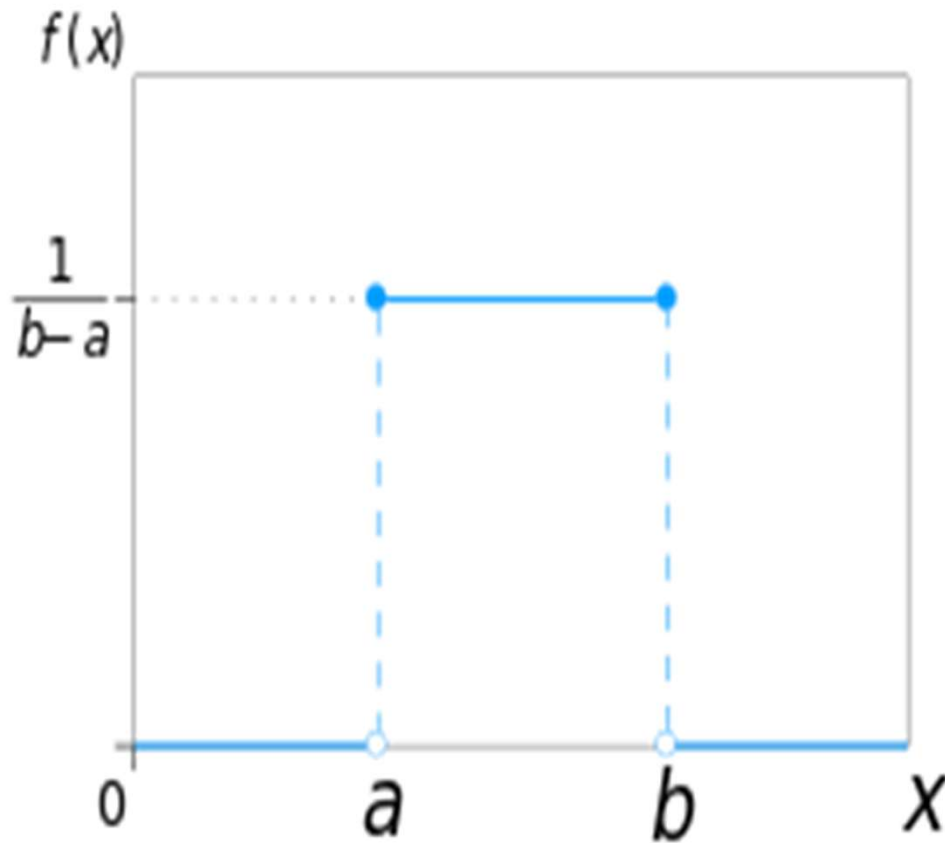
$$\begin{aligned} f_X(x) &= \lim_{\Delta x \rightarrow 0} \frac{P\{x \leq X \leq x + \Delta x\}}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{F_X(x + \Delta x) - F_X(x)}{\Delta x} = \frac{dF_X(x)}{dx} \end{aligned}$$



Probability Mass Function For Discrete Random Variables

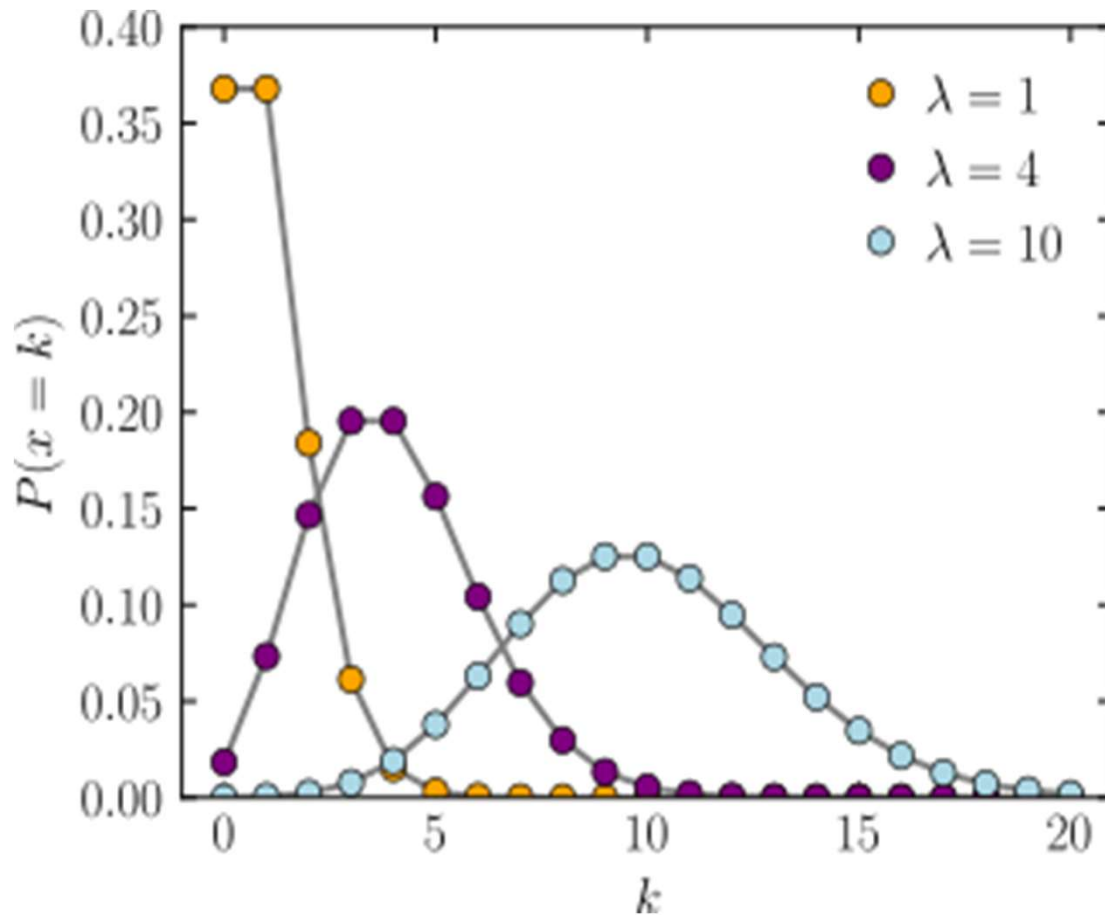


Uniform Distribution



$$x \sim U(x; a, b) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & x \notin [a, b] \end{cases}$$

Poisson Distribution

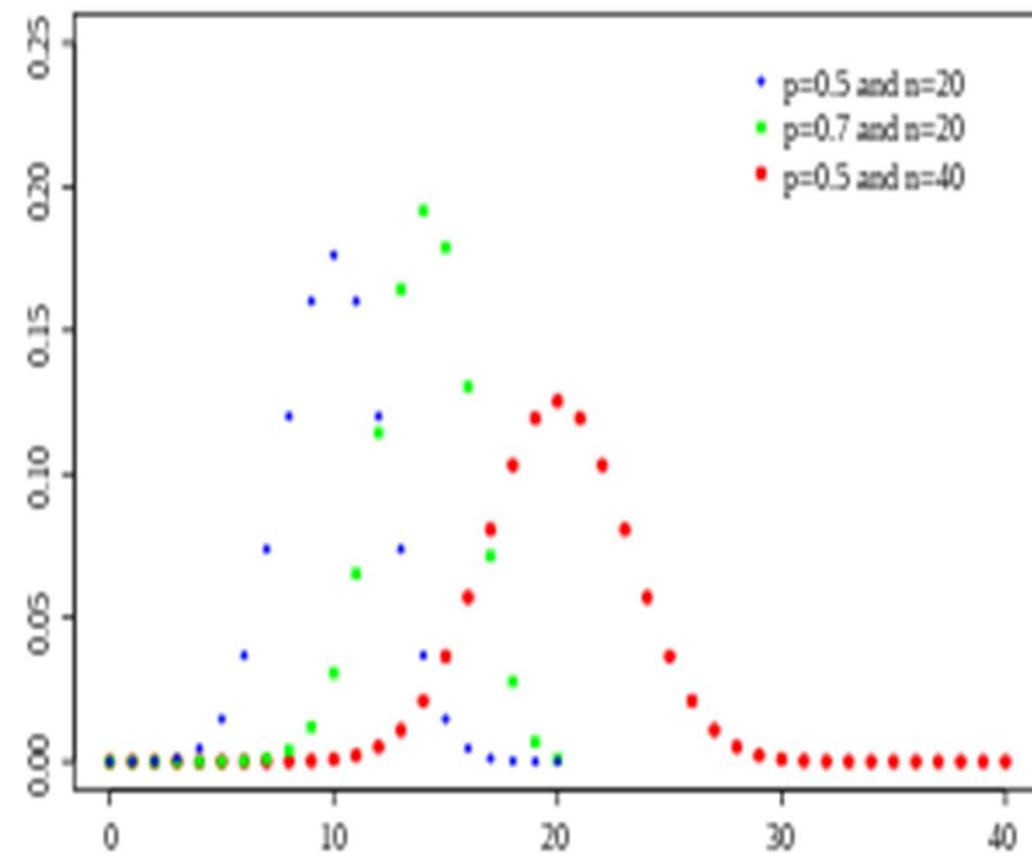


Discrete Probability Distribution of a given number of events (k) occurring in a fixed interval of time if these events occur with a known constant mean rate (λ) and independently of the time since the last event.

$$k \sim P(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

https://en.wikipedia.org/wiki/Poisson_distribution

Binomial Distribution



Probability Distribution of the number of successes (k) in a sequence of n independent experiments, each having a Boolean-valued outcome: success (with probability p) or failure (with probability $q = 1 - p$)

$$k \sim B(k; n, p) = \binom{n}{k} p^k q^{n-k}$$

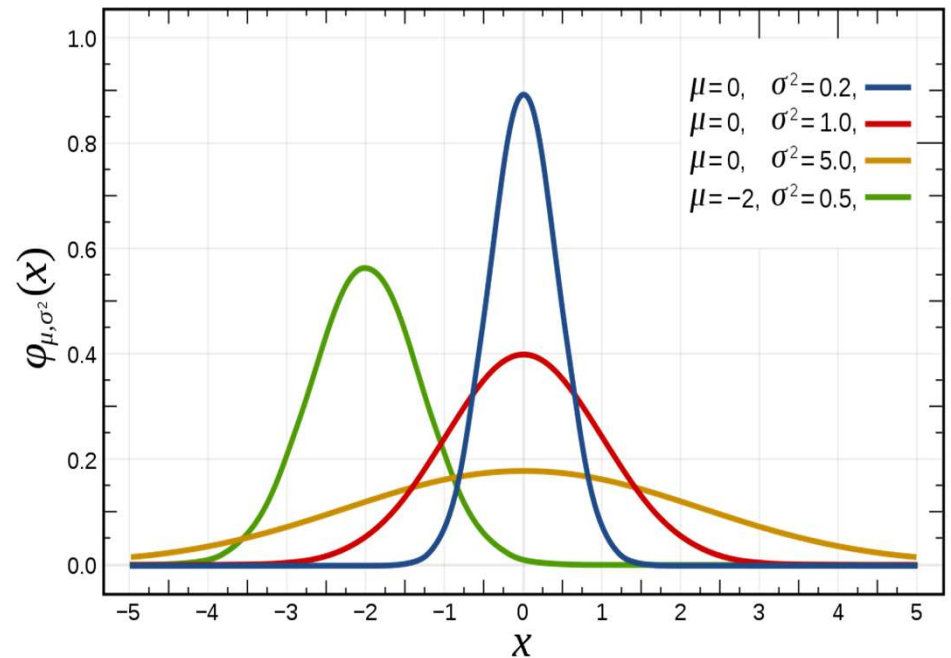
Gaussian Density Function

Single Variable

$$x \sim N(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Multi-Variable

$$x \sim N(x; \mu, \mathbf{C}) = \frac{1}{(2\pi)^{\frac{n}{2}} |\mathbf{C}|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \mathbf{C}^{-1} (x-\mu)}$$



Drawing a Random Number from Any P.M.F.

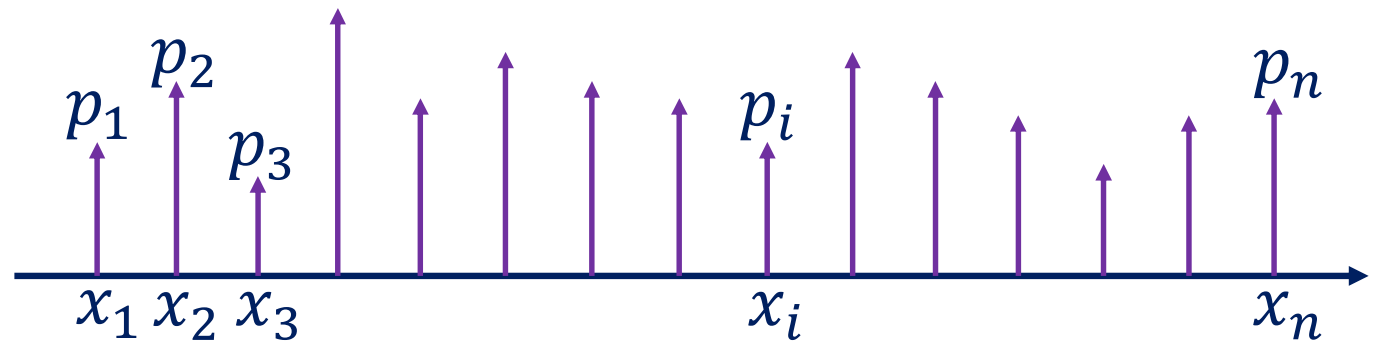
Input P.M.F.

$(x_i, p_i); i = 1, \dots, n$

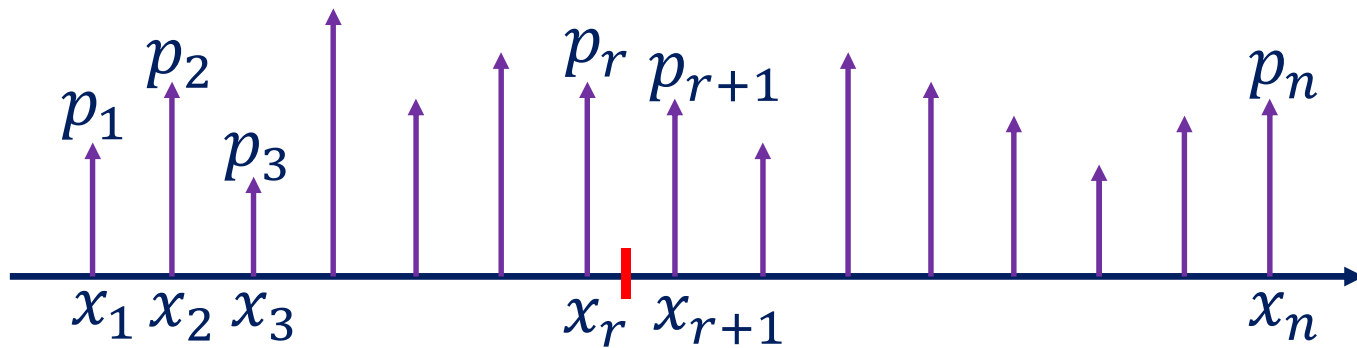
$$s_n = \sum_{i=1}^n p_i = 1$$

In Ascending
Order

$$s_m = \sum_{i=1}^m p_i$$



Drawing Random Number x from Any P.M.F.



$$p \sim U(0,1)$$

$$s_m = \sum_{i=1}^m p_i$$

$$r: s_r < p < s_{r+1}$$

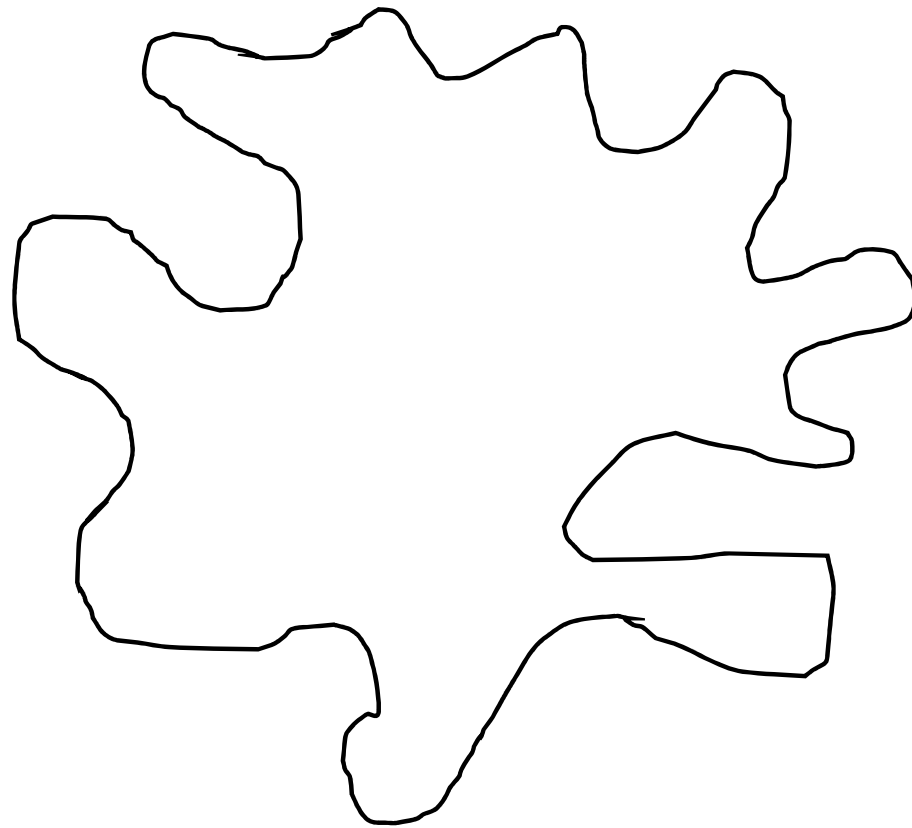
$$s_{r+1} = s_r + p_{r+1}$$

$$\frac{s_{r+1} - s_r}{x_{r+1} - x_r} = \frac{p - s_r}{x - x_r}$$

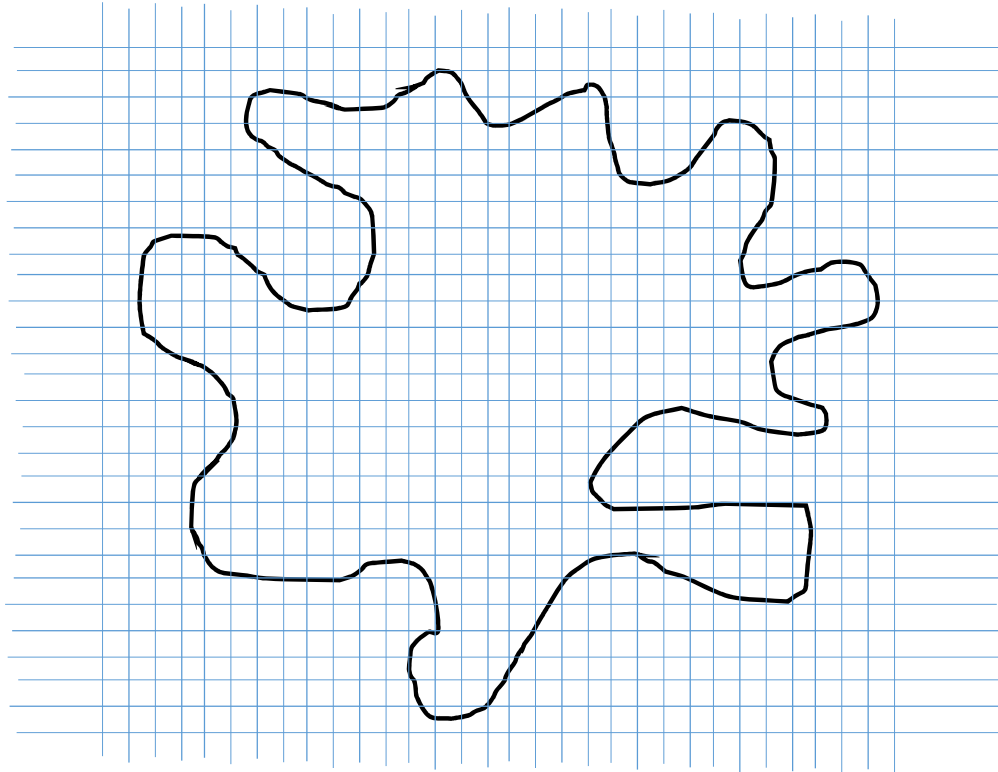


$$x = \frac{p - s_r}{p_{r+1}} (x_{r+1} - x_r)$$

How to Find the Area of this Region?

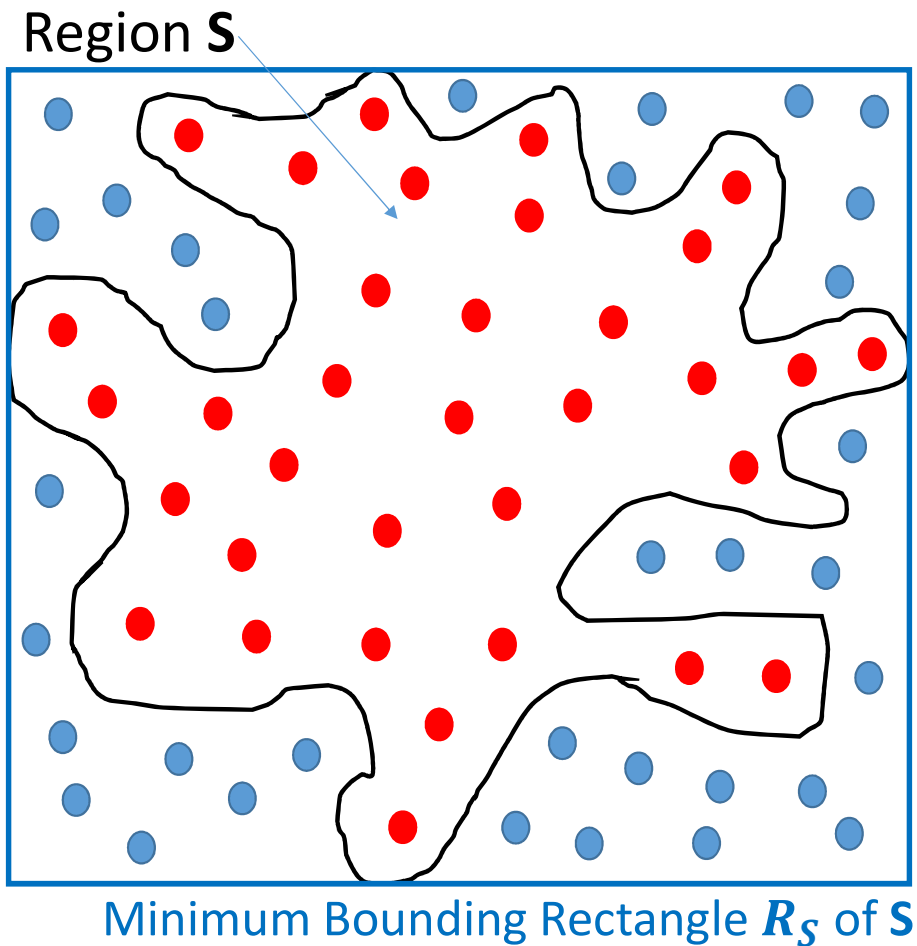


Calculus Approach



Decompose Region into Small Squares

Approximating Area by Random Sampling



N Number of Points are Sampled from a Uniform Distribution $U(R_S)$ Supported on R_S

- N_S Points within S
- $(N - N_S)$ Points within $R_S - S$

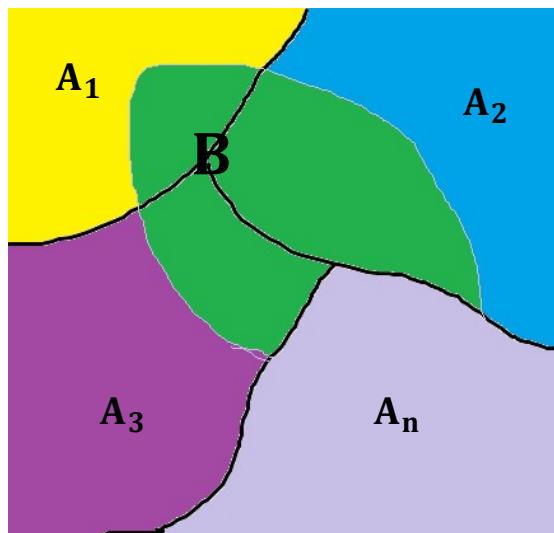
$$\hat{A}(S) = \left(\frac{N_S}{N} \right) A(R_S)$$

$$\lim_{N \rightarrow \infty} \hat{A}(S) = A(S)$$

Conditional Probability & Bayes Theorem

Conditional Probability: Event A assuming the Occurrence of B

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A|B)P(B) = P(A \cap B) = P(B|A)P(A)$$



Total Probability and Bayes Theorem

$$P(B) = P(B|A_1)P(A_1) + \cdots + P(B|A_n)P(A_n)$$

Conditionally Independent Events A & B

$$P(A \cap B|\lambda) = P(A|\lambda)P(B|\lambda)$$

Bayes Classifier

$$P(B) = P(B|A_1)P(A_1) + \cdots + P(B|A_n)P(A_n)$$

$$P(x) = P(x|\omega_1)P(\omega_1) + \cdots + P(x|\omega_n)P(\omega_n)$$

Feature
Vector

Classes or
Groups

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Likelihood

Prior

$$P(\omega_j|x) = \frac{P(x|\omega_j)P(\omega_j)}{P(x)}$$

Posterior

Evidence

Mathematical Expectation

Consider Random Variable x with Density/Mass Function $p(x)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(X)p(X)dX$$

$$E[h(X)] = \sum_i h(X_i)p(X_i)$$

Linear Operator: $E[a_1g(X) + a_2h(X)] = a_1E[g(X)] + a_2E[h(X)]$

Operating on a Constant K : $E[K] = K$

$$E[X]: \textit{Mean}$$

$$E[(X - \mu)^2]: \textit{Variance}$$

Example: Random Variable X with Poisson P.M.F.

$$P(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \Rightarrow \quad E(X) = \sum_{x=0}^{\infty} x P(x; \lambda) = \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda \cdot \lambda^{(x-1)}}{(x-1)!} = \lambda e^{-\lambda} \sum_{r=0}^{\infty} \frac{\lambda^r}{r!} = \lambda$$

$$E(X) = \mu_X = \lambda$$

Example: Random Variable X with Poisson P.M.F.

$$\text{Var}(X) = E[(X - \mu_X)^2] = ?$$

Example: Random Variable X with Poisson P.M.F.

$$\text{Var}(X) = E[(X - \mu_X)^2] = E[X^2 - 2X\mu_X + \mu_X^2]$$

$$\text{Var}(X) = E(X^2) - 2\mu_X E(X) + E(\mu_X^2) = E(X^2) - \mu_X^2$$

$$E(X^2) = \sum_{x=0}^{\infty} x^2 p(x) = \sum_{x=0}^{\infty} x(x-1)p(x) + \sum_{x=0}^{\infty} xp(x)$$

Example: Random Variable X with Poisson P.M.F.

$$\sum_{x=0}^{\infty} x(x-1)p(x) = \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^2 \cdot \lambda^{(x-2)}}{(x-2)!} = \lambda^2$$

$$\sum_{x=0}^{\infty} xp(x) = \lambda \quad \text{Var}(X) = E(X^2) - \mu_X^2 = \lambda^2 + \lambda - \lambda^2$$

$$\text{Var}(X) = \lambda$$

Markov Inequality

Consider the Random Variable $Y \geq 0$ and a Variable $a > 0$

$$P\{Y \geq a\} \leq \frac{E[Y]}{a}$$

Chebyshev Inequality

Substitutions $a = \lambda^2$ $\lambda > 0$

$$Y = \left(\frac{X - \mu}{\sigma} \right)^2$$

$\mu = E[X]$ (points to μ in the denominator)

$\sigma^2 = E[(X - \mu)^2]$ (points to σ in the denominator)

$E[Y] = 1$ (points to the entire fraction)

Chebyshev Inequality

Substituting in Markov Inequality

$$P\left\{\left(\frac{X - \mu}{\sigma}\right)^2 \geq \lambda^2\right\} \leq \frac{E[Y]}{\lambda^2}$$



$$P\{|X - \mu| \geq \lambda\sigma\} \leq \frac{1}{\lambda^2}$$

Chebyshev Inequality

Provides an Upper Bound on Membership for an Instance based on its Distance from Mean Normalized by Standard Deviation. No Properties of the Underlying Distribution are Assumed.

$$P\{|X - \mu| \geq \lambda\sigma\} \leq \frac{1}{\lambda^2}$$

Summary

- Probability: Basic Definitions and Postulates
- Random Variables, Distributions and Density Functions
- Conditional Probability, Independence and Bayes Theorem
- Application: Finding Area of an Irregular Shape
- Mathematical Expectation
- Chebyshev Inequality



Thank You

The images in this presentation are taken from different internet sources. These images are not created by Dr. Prithwijit Guha.